Lalith Koushik Vanam

+1 (281) 202-4868 · Houston, TX · lalithvanam04@gmail.com · linkedin.com/in/lalithvanam-dev · lalithkoushik.com

Professional Summary

Software Engineer with 4+ years of experience building production-grade web and agentic AI systems. Specialized in multi-agent orchestration with LangGraph and LangChain, full-stack development with ASP.NET Core and React, and RAG architectures over relational and graph databases. Track record of shipping maintainable, well-architected systems in industrial and AI-driven environments.

Technical Skills

Backend & APIs	ASP.NET Core MVC, Web API, FastAPI, RESTful Services
AI & LLM Systems	LangGraph, LangChain, Multi-Agent Orchestration, Claude (Haiku/Sonnet), RAG Pipelines, SQL-RAG, Graph-RAG (NetworkX), Ollama, Conversation Memory
Agent Patterns	Supervisor-Worker Orchestration, Pre-Execution Planning, Human-in-the-Loop, Tool Result Caching, Tenacity Retry Policies, Model Tiering
Frontend	React.js, TypeScript, Plotly.js, JavaScript, HTML/CSS
Databases	PostgreSQL, TimescaleDB, MSSQL, Heidi SQL
Architecture	Clean Architecture, Repository Pattern, Microservices, API-first Design
DevOps & Tooling	CI/CD Pipelines, Git, Cloudflare, asyncpg, httpx

Professional Experience

Full Stack & AI Software Engineer — June 2024 – May 2026
MidstreamAI — *Houston, TX*

- **Multi-agent platform architecture:** Designed and owned a production-grade multi-agent AI system using LangGraph and LangChain — Supervisor → Specialist Agents (Topology, Operations) → Summarizer — enabling natural-language queries over industrial infrastructure with full audit logging via a shared Pydantic V2 `AgentState` as the single source of truth.
- **Two-stage Supervisor design:** Built a Supervisor agent with a classification stage (6 routing strategies: *topology, operations, topology_first, operations_first, parallel, summarizer*) followed by a pre-execution planning stage that decomposes queries into dependency-aware agent tasks entirely upfront — no mid-flight replanning required.
- **Human-in-the-loop clarification:** Engineered an ambiguity detection flow where genuinely unclear queries return a structured clarification question with selectable options rendered as UI buttons; specialist agents are bypassed until the operator confirms intent.
- **Custom LangGraph mini-workflow:** Replaced `create_react_agent` with a hand-wired 3-node graph (*call_llm → execute_tool → handle_response*) to gain full control over response normalization into typed structures (*devices[], paths[][], metadata{}*), structured error recovery distinguishing terminal vs. recoverable failures, and iteration capping to bound cost.
- **Production resilience:** Implemented in-memory tool result caching to eliminate redundant HTTP round-trips; Tenacity exponential backoff retry (2s–30s, 3 attempts) for transient LLM errors (429/5xx/timeouts); deliberate model tiering — Claude Haiku for tool-selection tasks (10× higher TPM, 20× lower cost per token) and Sonnet for final response synthesis.
- **Full-stack platform delivery:** Designed and delivered a pipeline leak detection analytics platform (ASP.NET Core MVC, Plotly.js) with interactive dashboards and server-side aggregation; built reusable frontend and backend components adopted across multiple enterprise applications.

Full Stack Developer — August 2022 – May 2024
LTIMindtree — *Hyderabad, India*

- Developed and maintained industrial seep detection and leak analysis systems for real-world oil and gas pipeline monitoring, working across the full software lifecycle from configuration generation to production delivery.
- Redesigned seep plot and alarm analysis workflows, integrating trend-based visualizations via Excel macros and dataPARC systems to improve operational insight for field engineers.
- Collaborated with domain experts to validate alarm logic, ensure data accuracy, and meet reliability standards for safety-critical infrastructure software used in live operational environments.
- Gained end-to-end exposure to enterprise industrial software: real-time data ingestion, sensor-driven visualization, configuration management, and production validation in high-stakes environments.

Selected Projects

Agentic-AI Analytics Platform *FastAPI · LangGraph · LangChain · Claude · PostgreSQL/TimescaleDB · NetworkX · React/TypeScript*

- **Key architectural decision — custom graph over `create_react_agent`:** Built a manual 3-node LangGraph mini-workflow with a dedicated `handle_response` node between tool execution and the next LLM call. This enables response normalization, recoverable error hints (device not found → fuzzy search → retry with canonical ID), and hard stops on terminal errors — none of which are possible inside `create_react_agent`'s fixed wiring.

- **Conversation memory and multi-turn resolution:** Integrated session-level conversation memory into both the Supervisor (for pronoun/reference resolution across turns) and the Summarizer (for contextually accurate follow-up answers), moving the system from stateless single-turn responses to coherent multi-turn dialogue.
- **SQL-RAG and Graph-RAG agents:** Implemented an Operations Agent querying TimescaleDB via MCP-based tool discovery for time-series sensor data, and a Topology Agent performing graph traversal over a NetworkX-modelled pipeline network — both feeding results into a shared AgentState for unified summarization.

Pipeline Leak Detection Insights Application *ASP.NET Core MVC · Plotly.js · PostgreSQL*

- Built a production-ready analytics application for pipeline leak detection: interactive Plotly.js dashboards, repository-pattern data access, server-side aggregation controllers, and dynamic client-side filtering — designed for scalability and long-term maintainability in an industrial environment.

Education

University of Houston
Master of Science in Computer Science
Keshav Memorial Institute of Technology
Bachelor of Technology in Computer Science